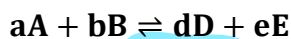


Chapter 15 Chemical Equilibrium

Review

- **Chemical equilibrium:** occurs when opposing reactions proceed at equal rates: the rate at which the products form from the reactants equals the rate at which the reactants form from the products.
 - 1) Only the reversible reaction in closed system which under a constant temperature can establish a chemical equilibrium;
 - 2) Chemical equilibrium is a dynamic equilibrium, the forward reaction rate equals the reverse reaction rate;
 - 3) At equilibrium, the concentration of reactants and products no longer change with time;
 - 4) For equilibrium to occur, neither reactants nor products can escape from the system;
 - 5) At equilibrium, a particular ratio of concentration terms equals a constant.

- **Equilibrium-constant:**



$$K_c = \frac{[D]^d [E]^e}{[A]^a [B]^b}$$

- **K_c : concentration equilibrium constant.**

- 1) The equilibrium-constant expression depends only on the stoichiometry of the reaction, not on its mechanism;
- 2) The value of K_c depends only on the particular reaction and on the temperature;

- **K_p : pressure equilibrium constant.**

$$K_p = \frac{[P_D]^d [P_E]^e}{[P_A]^a [P_B]^b}$$

Diff Partial pressure Distancing Q/K
P_{sub} values

- **Relationship between K_c and K_p :**

$$K_p = K_c (RT)^{\Delta n}$$

$$\Delta n = (\text{moles of gaseous product}) - (\text{moles of gaseous reactant})$$

- If $K \gg 1$ (large K): equilibrium lies to right, products predominate;
If $K \ll 1$ (small K): equilibrium lies to left, reactants predominate;

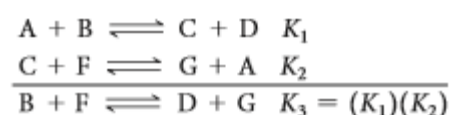
- 1) The equilibrium constant of a reaction in the reverse direction is the inverse of the equilibrium constant of the reaction in the forward direction:



- 2) The equilibrium constant of a reaction that has been multiplied by a number is equal to the original equilibrium constant raised to a power equal to that number:



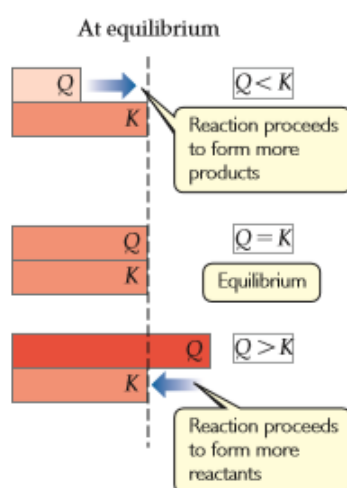
- 3) The equilibrium constant for a net reaction made up of two or more reactions is the product of the equilibrium constants for the individual reactions:



- **Homogeneous equilibria:** equilibria involve substances that are all in the same phase.
- **Heterogeneous equilibria:** equilibria involve substances that are in different phases.
 - 1) Whenever a solid or pure liquid is involved in a heterogeneous equilibrium, its concentration isn't included in the equilibrium-constant expression;
 - 2) The concentration of solvent in a dilute solution isn't included in the equilibrium-constant expression;
 - 3) In a non-aqueous system, the concentration of water should be in the equilibrium constant expression;
- **Reaction quotient (Q):** a number obtained by substituting reactant and product concentrations or partial pressures at any point during a reaction into an equilibrium-constant expression.

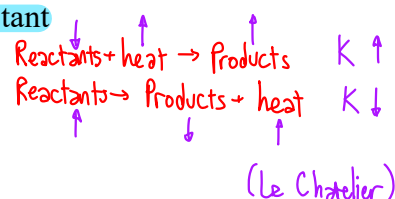
$$\begin{array}{c}
 \mathbf{aA + bB \rightleftharpoons dD + eE} \\
 Q_c = \frac{[D]^d[E]^e}{[A]^a[B]^b} \quad Q_p = \frac{[P_D]^d[P_E]^e}{[P_A]^a[P_B]^b}
 \end{array}$$

- 1) $Q < K$: The concentration of products is too small and that of reactants too large. The reaction achieves equilibrium by forming more products; it proceeds from left to right.
- 2) $Q = K$: The reaction quotient equals the equilibrium constant only if the system is at equilibrium.
- 3) $Q > K$: The concentration of products is too large and that of reactants too small. The reaction achieves equilibrium by forming more reactants; it proceeds from right to left.



ICE Table

- **Le Chatelier's principle:** If a system at equilibrium is disturbed by a change in temperature, pressure, or a component concentration, the system will shift its equilibrium position so as to counteract the effect of the disturbance.
 - ✓ Changing pressure, or concentration K value keeps constant
 - ✓ Endothermic Increasing T results in higher K value
 - ✓ Exothermic Increasing T results in lower K value



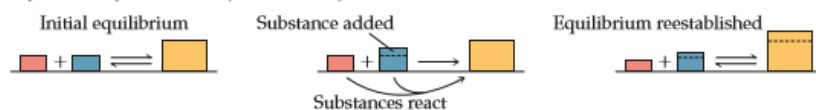
$P \uparrow \rightarrow \# \text{moles} \downarrow$ (if two sides have not equal #moles of gas)

Le Châtelier's Principle

If a system at equilibrium is disturbed by a change in **concentration, pressure, or temperature**, the system will shift its equilibrium position so as to counter the effect of the disturbance.

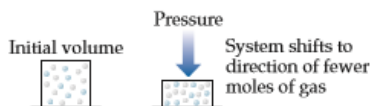
Concentration: adding or removing a reactant or product

If a substance is added to a system at equilibrium, the system reacts to consume some of the substance. If a substance is removed from a system, the system reacts to produce more of substance.



Pressure: changing the pressure by changing the volume

At constant temperature, reducing the volume of a gaseous equilibrium mixture causes the system to shift in the direction that reduces the number of moles of gas.



Temperature:

If the temperature of a system at equilibrium is increased, the system reacts as if we added a reactant to an endothermic reaction or a product to an exothermic reaction. The equilibrium shifts in the direction that consumes the "excess reactant," namely heat.



- **Catalyst:** a catalyst **increases the rate** at which equilibrium is achieved but **does not change the composition of the equilibrium mixture.**